

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
Faculty Name	Dr. Sukanya
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Branch	CSE
Course Outcome	6

Case Study Topic: Day-Trading Optimal Strategy using Dynamic Programming (Best Time to Buy and Sell Stock IV)

Work Done Summary:

In this case study, Dynamic Programming was used to solve the stock trading optimization problem where a trader can perform at most **k transactions**.

Implementation Details:

1. Problem Definition
2. Dynamic Programming Approach
3. DP Table Construction
4. Optimal Transaction Identification
5. Complexity Analysis

Result:

Unlimited Transactions :

Maximum Profit :

120

At Most Two Transactions :

Maximum Profit :

85

Best Trading Plan :

Buy @ 80

Sell @ 110

Buy @ 90

Sell @ 145

Output:

```
Profit for  $k = 2$  : 90
Profit for  $k = \infty$  : 120
```

GitHub Link:

<https://github.com/saisharan134/DSA2-Case-Studies>

Student Signature	Faculty Signature